

★ PRIMER · UNIT 5

Before the Lecture: From a Number to a Verdict

Unit 5 · Classification and Anomalies

Watch this first if probability, odds and distance are hazy

Twenty minutes of groundwork — how a number becomes a probability, how you measure distance between two customers, and the one rule behind the diagnosing doctor.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Sigmoid and distance are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 5 Sits

The course builds one pipeline; here is this unit's stone

Unit 4

done

The straight line, and how to tame it.

Unit 5

you are here

Turning a score into a yes or no.

Unit 6

next

The widest road — support vectors.

Regression predicts how much. Classification predicts which one. The tools differ.

What Classification Changes

Regression answers how much. Classification answers which one — and the output must be a probability.

A regression output

- Any number at all.
- -3.2 is perfectly legal.
- Nothing bounds it.

A classifier output

- Must sit between 0 and 1.
- It is a probability.
- Something must squash it.

That squashing function is the sigmoid, and it is the whole of logistic regression.

The Words the Lecture Assumes

Five terms, and all three classifiers open up

Word	What it means	Here
Probability	a number from 0 to 1	chance this loan defaults
Odds	probability over one minus it	0.75 becomes 3 to 1
Sigmoid	the S-curve that squashes	turns a score into a chance
Threshold	where you cut yes from no	0.5 is a choice, not a law
Euclidean distance	straight-line gap between points	how alike two customers are

Odds and probability carry the same information. Logistic regression is linear in odds.

Three Classifiers, Three Philosophies

The lecture takes each in turn; they disagree about what a model is for

Method	How it decides	What it needs
Logistic	fits an S-curve to a score	a shape assumption
KNN	asks the nearest neighbours	a distance, and all the data
Naive Bayes	updates a belief with evidence	a prior, and independence

Discriminative, instance-based, generative. Three answers to one question.

Why 0.5 Is a Choice, Not a Law

The threshold is a business decision. Move it and you trade false alarms against missed frauds.

Lower the threshold

- You flag more cases.
- Recall rises.
- False alarms rise too.

Raise it

- You flag only the certain.
- Precision rises.
- You miss real fraud.

The model gives a probability. Where to cut it is not the model's business.

The S-Curve by Hand

Five scores through the sigmoid — the shape is the point

Score z	e to the $-z$	1 + that	Probability
-2	7.39	8.39	0.12
-1	2.72	3.72	0.27
0	1.00	2.00	0.50
+1	0.37	1.37	0.73
+2	0.14	1.14	0.88

A score of zero is always a coin flip. The curve is symmetric about that point.

Distance Between Two Customers

KNN needs a number for 'alike' — this is that number

Two customers

- A: age 30, income 40.
- B: age 34, income 43.
- Two features each.

The gap

- Differences: 4 and 3.
- Square, add: $16 + 9 = 25$.
- Square root: distance is 5.

Scale your features first — income in won would swamp age entirely

Why Naive Bayes Is Called Naive

It makes an assumption it knows is false, and works anyway

The assumption

- Every clue is independent.
- Income tells nothing of debt.
- This is plainly untrue.

Why it survives

- It only ranks cases.
- Ranking is robust to the lie.
- It is fast, and a fine filter.

A wrong assumption that produces a useful ordering is a bargain, not a scandal.

How a Belief Gets Updated

Bayes' rule in three moves — the lecture works it in full

1 · Start

the prior

Before any evidence, 3% of loans default. That is your starting belief.

2 · Weigh

the evidence

This applicant's income is far more common among defaulters.

3 · Update

the posterior

The belief climbs. Add another clue and it climbs again.

The prior matters most when evidence is thin — which in fraud is most of the time.

Four Anomaly Detectors, One Thread

The course meets one per unit; this is the map you will keep returning to

Detector	Unit	Idea
LOF	5 · this one	is this point in a sparse neighbourhood?
One-class SVM	6	draw a boundary around the normal
Isolation Forest	7	how few splits to isolate this point?
Autoencoder	8	can the network rebuild this point?

Each catches what the last one missed. In production they are stacked, not chosen.

Notation You Will Meet

Four symbols carry the classification section

Symbol	Read it as	It stands for
$\sigma(z)$	sigma of z	the sigmoid — squashes to 0 and 1
$P(A B)$	P of A given B	the chance of A once B is known
k	kay	how many neighbours KNN asks
$d(a,b)$	distance a b	how far apart two points are

The vertical bar in $P(A|B)$ means 'given'. It is the whole of Bayes' rule.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>LogisticRegression().fit(X, y)</code>	fit the S-curve
<code>model.predict_proba(X)</code>	probabilities, not yes-or-no verdicts
<code>KNeighborsClassifier(n_neighbors=5)</code>	ask the five nearest
<code>StandardScaler().fit_transform(X)</code>	put features on one scale — KNN needs this
<code>LocalOutlierFactor()</code>	the unit's anomaly detector

Use `predict_proba`, not `predict`. The threshold is yours to set, not the library's.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 6

1

"The model decides at 0.5"

0.5 is a default someone chose. In fraud detection the right cut is often far lower.

2

"KNN needs no training, so it is cheap"

It does no training and all its work at prediction time, against every stored point.

3

"Naive Bayes is broken because the assumption is false"

The assumption is false and the ranking still works. Judge a model by its output.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1 A score of 0 goes through the sigmoid. What comes out?

0.50

2 Gaps of 4 and 3. What is the Euclidean distance?

5

3 Which classifier must have its features scaled first?

KNN

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It builds the S-curve properly

From a banker's gut feel to a probability, with the curve worked by hand.

2

It contrasts three philosophies

Logistic, KNN and naive Bayes — discriminative, instance-based, generative.

3

It ends in code

The problem set runs logistic by hand, a KNN vote, and LOF on the odd customer out.

Next: Unit 5 lecture — Classification and Our First Anomaly Detector